

# PBL Question 9: JavaScript Inheritance

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## Aim

To study and implement Single, Multilevel, Hierarchical and Multiple Inheritance in JavaScript.

## Theory

Inheritance allows a class to acquire properties and methods from another class. JavaScript supports Single, Multilevel and Hierarchical Inheritance using classes and extends. Multiple inheritance can be simulated using Mixins.

## Program

```
[Program]

// SINGLE INHERITANCE
class Animal {
  constructor(name) {
    this.name = name;
  }
  eat() {
    console.log(`${this.name} is eating.`);
  }
}

class Dog extends Animal {
  bark() {
    console.log(`${this.name} is barking.`);
  }
}

const myDog = new Dog("Rocky");
myDog.eat();
myDog.bark();

// MULTILEVEL INHERITANCE
class Vehicle {
  move() {
    console.log("Moving forward.");
  }
}

class Bike extends Vehicle {
  wheels() {
    console.log("Bike has 2 wheels.");
  }
}

class Car extends Bike {
  windows() {
    console.log("Car has 4 windows.");
  }
}

// HIERARCHICAL INHERITANCE
class User {
  constructor(username) {
    this.username = username;
  }
  login() {
    console.log(`${this.username} logged in.`);
  }
}

// MULTIPLE INHERITANCE USING MIXINS
Object.assign(Area.prototype, Hall, Bedroom);
```

## Expected Output

Rocky is eating.  
Rocky is barking.  
Moving forward.  
Bike has 2 wheels.  
Car has 4 windows.  
Karthik logged in.  
Ram logged in.  
Karthik deleted a user.  
Ram checked out successfully.  
Building House No: 101  
Area: MVP Colony  
Sofa is placed in Hall.  
TV is placed in Hall.  
Bed is in Bedroom.  
Closet is in Bedroom.

## Result

The program was executed successfully and demonstrated Single, Multilevel, Hierarchical and Multiple Inheritance in JavaScript.